Capped GBM pricing

Replication, numerical convergence and Monte Carlo sensitivities

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11 October 2026

Abstract

I connect no-arbitrage replication, martingale valuation and numerical pricing for a capped European claim under geometric Brownian motion with continuous dividends. A heat-coordinate transformation permits a monotone backward-Euler scheme with unconditional maximum-norm stability. I prove convergence for the actual kinked payoff and separate finite-domain, interpolation and discretisation errors. Explicit central-difference bounds establish joint price and Greek consistency when domains, meshes and bumps are refined appropriately. Exact terminal simulation supplies pathwise first sensitivities and likelihood-ratio gamma. A mixed sensitivity derivation gives a bounded gamma estimator with finite-sample concentration. Seven synthetic contracts compare analytic, PDE and Monte Carlo prices and five Greeks at declared sampling budgets. The mixed gamma estimator lowers measured raw variance, while independently fitted stock controls worsen some sensitivities. Detailed derivations and adverse numerical findings are retained. The analytic call spread remains the strongest valuation baseline; this companion demonstrates and checks the financial and computational foundations supporting the basket-pricing project.

Keywords: derivative pricing; geometric Brownian motion; replication; finite differences; numerical stability; convergence; Monte Carlo Greeks.

JEL classification: C15, C63, G13.

1 Financial model and no-arbitrage valuation

I use a one-asset claim to connect replication, martingale valuation, parabolic equations and statistical estimation under one financial model. The purpose is to demonstrate the mathematics underlying the numerical comparisons. The classical valuation framework follows Black and Scholes (1973) and Merton (1973); these established methods do not establish a quantum advantage or a new financial theory. Replication and call valuation specialise Black and Scholes (1973); the continuous-dividend treatment follows Merton (1973). The pricing and sensitivity derivations below retain the assumptions required for these specialisations.

Let $S > 0$, $K > 0$, $H > 0$, $\sigma > 0$ and $\tau > 0$. The interest rate r and continuous dividend yield q are constant. Write $J = K + H$ and

$$F(x) = \min\{(x - K)^+, H\} = (x - K)^+ - (x - J)^+, \qquad D = \exp(-r\tau), \qquad v = \sigma\sqrt{\tau}. \tag{1}$$

Under the selected pricing measure Q, the ex-dividend asset follows

$$dS_t = (r - q)S_t\,dt + \sigma S_t\,dW_t^Q, \qquad X = S\exp\{b\tau + vZ\}, \qquad b = r - q - \tfrac{1}{2}\sigma^2, \tag{2}$$

where Z is standard normal. The price is

$$V(S, \tau; r, q, \sigma) = D\,\mathbb{E}_Q[F(X)]. \tag{3}$$

This is a model comparison with synthetic parameters, rather than a calibrated empirical test. A physical return mean cannot be substituted for the pricing drift $r - q$. I assume frictionless

continuous trading, reinvestable dividends, borrowing and lending at the same rate r, and an unrestricted stock position within the stated admissible strategy class. The probability space carries the usual augmented Brownian filtration, satisfying the usual completeness and right-continuity conditions.

Proposition 1.1 (Replication and the pricing equation)**.** *The price in* (3) *is bounded, continuous at expiry and smooth for $S, \tau > 0$. It satisfies*

$$V_\tau = \tfrac{1}{2}\sigma^2 S^2 V_{SS} + (r-q)SV_S - rV, \qquad V(S,0) = F(S). \tag{4}$$

Holding V_S shares and placing $V - SV_S$ in the money-market account replicates the claim when dividends are reinvested within the self-financing strategy. Its discounted wealth is non-negative, and its stochastic gain is square-integrable on each finite horizon. Under the usual admissibility condition that discounted wealth is bounded below, the stipulated market has no arbitrage. This replicating price is the unique no-arbitrage price among bounded claim valuations and the unique bounded classical solution of the terminal-value problem. Here a classical solution is $C^{1,2}$ for positive remaining maturity and is continuous at expiry.

The dividend term belongs in the self-financing gain equation, not in the terminal payoff. Appendix A derives both the measure change and the replication argument, including the limiting argument at the payoff kinks.

2 Analytic prices and Greeks

Let Φ and φ denote the standard normal distribution function and density. For strike $A > 0$, define

$$d_{1,A} = \frac{\log(S/A) + (r - q + \sigma^2/2)\tau}{v}, \qquad d_{2,A} = d_{1,A} - v, \qquad C_A = Se^{-q\tau}\Phi(d_{1,A}) - AD\Phi(d_{2,A}). \tag{5}$$

Then $V = C_K - C_J$. Write $\Delta\Phi_i = \Phi(d_{i,K}) - \Phi(d_{i,J})$ and $\Delta\varphi = \varphi(d_{1,K}) - \varphi(d_{1,J})$. Differentiation gives

$$\Delta = V_S = e^{-q\tau}\Delta\Phi_1, \tag{6}$$

$$\Gamma = V_{SS} = \frac{e^{-q\tau}\Delta\varphi}{Sv}, \tag{7}$$

$$\mathcal{V} = V_\sigma = Se^{-q\tau}\sqrt{\tau}\,\Delta\varphi, \tag{8}$$

$$\mathcal{R} = V_r = \tau D\{K\Phi(d_{2,K}) - J\Phi(d_{2,J})\}, \tag{9}$$

$$\Theta = -V_\tau = -\frac{Se^{-q\tau}\sigma}{2\sqrt{\tau}}\Delta\varphi + qSe^{-q\tau}\Delta\Phi_1 - rD\{K\Phi(d_{2,K}) - J\Phi(d_{2,J})\}. \tag{10}$$

Vega and rho are derivatives per unit change in volatility and interest rate. Calendar theta uses a fixed expiry and advances the valuation date, so it is the negative maturity derivative. Gamma and vega can be negative for a capped claim; this is a property of the call spread, not a numerical failure.

3 A monotone heat-equation discretisation

Introduce $y = \log(S/K) + b\tau$, $a = \sigma^2/2$ and $f(y) = F(Ke^y)$. The whole-line heat solution w satisfies

$$w_\tau = aw_{yy}, \qquad w(0,y) = f(y), \qquad V(S,\tau) = Dw(\tau, y). \tag{11}$$

Its payoff kinks lie at 0 and $c = \log(J/K)$. Moreover, $0 \le f \le H$ and f is globally Lipschitz with constant $L_f = J$. For a finite domain $L < 0 < c < U$, let w^{box} solve the same equation with fixed boundary values 0 and H. These values approximate the exact financial far-field limits

$$\lim_{S\downarrow 0} V(S,\tau) = 0, \qquad \lim_{S\to\infty} V(S,\tau) = DH. \tag{12}$$

Indeed the terminal asset tends pointwise to zero or infinity as its initial spot does, and the bounded continuous payoff permits dominated convergence. At remaining maturity τ, the heat-domain endpoints correspond to asset values $K\exp(L - b\tau)$ and $K\exp(U - b\tau)$. Their finite values do not make the imposed boundary prices exact; the exit-tail bound below quantifies the approximation.

For integers $M \ge 3$ and $N \ge 1$, set $h = (U - L)/M$, $y_j = L + jh$, $k = \tau/N$ and $\lambda = ak/h^2$. Backward Euler gives

$$-\lambda W_{j-1}^n + (1 + 2\lambda)W_j^n - \lambda W_{j+1}^n = W_j^{n-1}, \qquad W_0^n = 0, \quad W_M^n = H, \quad W_j^0 = f(y_j). \tag{13}$$

I interpolate the final grid values linearly at the query y and then apply the discount factor. Subtracting the affine boundary lift diagonalises this same finite-difference operator in the discrete sine basis. Its eigenvalues and backward-Euler factors are

$$\nu_\ell = \frac{4}{h^2}\sin^2\left(\frac{\ell\pi}{2M}\right), \qquad (1 + ak\nu_\ell)^{-N}, \qquad 1 \le \ell \le M - 1. \tag{14}$$

This is a faster evaluation of (13), rather than a different spatial method. A direct tridiagonal time-step implementation checks that equivalence numerically. A floating-point implementation is not an interval-arithmetic certificate merely because its mathematical scheme has a proved bound.

Theorem 3.1 (Unconditional stability and a nonsmooth-payoff error bound)**.** *For every $h, k > 0$,* (13) *preserves $0 \le W_j^n \le H$. Differences between two solutions with the same boundary values contract in the maximum norm at every step. Fix an even, non-negative $\rho \in C_c^\infty(-1,1)$ with integral one, and set*

$$m_1 = \int_{-1}^{1} |z|\rho(z)\,dz, \qquad c_3 = \|\rho'''\|_{L^1}, \qquad 0 < \eta < \min\{-L, U - c\}. \tag{15}$$

For exact arithmetic, the nodal error at maturity obeys

$$\max_{1\le j\le M-1} |W_j^N - w^{\mathrm{box}}(\tau, y_j)| \le 2L_f m_1 \eta + \tau\left(\frac{ah^2}{12} + \frac{a^2k}{2}\right) L_f c_3 \eta^{-3}. \tag{16}$$

Consequently the scheme converges to the finite-domain solution, with a rigorous upper bound of order $(h^2 + k)^{1/4}$ for this nonsmooth payoff on a fixed time interval. This conservative bound does not assert that the observed rate is exactly one quarter.

The proof uses the maximum principle, a compact mollifier and an odd periodic extension after the boundary lift. It does not assume that the unmollified payoff has four derivatives. The full proof is in Appendix B.

Proposition 3.2 (Domain and interpolation errors)**.** *For $x \in [L, U]$, put*

$$\beta(\tau, x) = \min\left\{1, 2\Phi\left(-\frac{x - L}{\sigma\sqrt{\tau}}\right) + 2\Phi\left(-\frac{U - x}{\sigma\sqrt{\tau}}\right)\right\}. \tag{17}$$

Then $|w(\tau,x) - w^{\mathrm{box}}(\tau,x)| \le H\beta(\tau,x)$. *Let* y *lie between grid nodes* y_j, y_{j+1} *and define* $\beta_* = \max\{\beta(\tau,y_j), \beta(\tau,y_{j+1})\}$. *The interpolated, discounted price* $\widehat{V}$ *satisfies*

$$|\widehat{V} - V| \le D\left[H\beta_* + 2L_f m_1 \eta + \tau\left(\frac{ah^2}{12} + \frac{a^2 k}{2}\right) L_f c_3 \eta^{-3} + \frac{h^2 L_f}{8\sigma\sqrt{\tau}}\sqrt{\frac{2}{\pi}}\right]. \tag{18}$$

The four terms control the finite domain, payoff smoothing, numerical consistency and interpolation. Payoff smoothing is a proof device; the implemented initial condition remains the original capped payoff. On a fixed domain, mesh refinement converges to w^{box}, leaving a domain error relative to the whole-line financial price. Full-price consistency requires the domain error to vanish as well.

4 Greek consistency and Monte Carlo comparisons

Nonsmooth payoffs require separate control of sensitivities; the analysis of Giles and Carter (2006) gives a published comparison for Crank–Nicolson/ Rannacher schemes. I prove the bounds for the backward-Euler scheme used here.

I compute PDE Greeks by central price differences on the same domain and grid. This fixes the numerical heat domain under a bump; the transformed query y, diffusivity a and discount factor still change with the bumped parameter. The small-bump limit must respect the price error.

Proposition 4.1 (Propagation into central differences)**.** *Suppose* $|\widehat{V} - V| \le E$ *at every required bumped point. For a scalar parameter* p, *if* $|\partial_p^3 V| \le M_3$ *on its bump interval, then*

$$\left|\frac{\widehat{V}(p+d) - \widehat{V}(p-d)}{2d} - \partial_p V(p)\right| \le \frac{E}{d} + \frac{M_3 d^2}{6}. \tag{19}$$

For spot gamma, if $|\partial_S^4 V| \le M_4$ *on the spot bump interval, then*

$$\left|\frac{\widehat{V}(S+d) - 2\widehat{V}(S) + \widehat{V}(S-d)}{d^2} - \Gamma\right| \le \frac{4E}{d^2} + \frac{M_4 d^2}{12}. \tag{20}$$

These statements apply uniformly on a compact parameter set with S, σ, τ *bounded away from zero, provided all queries lie inside the domain. Refining the mesh and enlarging the domain so that* $E_n \to 0$, *then choosing* $d_n \to 0$ *and* $E_n/d_n \to 0$ *for first derivatives or* $E_n/d_n^2 \to 0$ *for gamma, proves consistency for the whole-line Greeks.*

After fixing units, choices proportional to $E_n^{1/3}$ and $E_n^{1/4}$ balance the displayed error terms for first derivatives and gamma, respectively. Practical comparisons also retain the exact analytic central-difference bias at each implemented bump. This diagnoses numerical price error separately from bump bias and prevents an apparent Greek discrepancy from being assigned wholly to the PDE scheme.

The implemented bump recipe uses a dimensionless log-grid spacing h, a positive spot reference scale S_*, and positive reference scales for the other parameters. The spot bump is proportional to $S_* h^{1/8}$; first-order parameter bumps are proportional to $h^{1/4}$ in their respective reference units. With $k = O(h^2)$, the rigorous price bound is $O(h^{1/2})$ after making the domain error $o(h^{1/2})$. The resulting spot gamma error is $O(h^{1/4})$, and the first-order errors are at most $O(h^{1/4})$. These are conservative guaranteed upper rates, not fitted empirical orders. A finite fixed domain retains its displayed exit-tail allowance, regardless of this bump recipe.

Pathwise and likelihood-ratio estimators have established foundations (Broadie and Glasserman, 1996). The capped-payoff formulae and their ranges are derived explicitly in Appendix C.

For exact terminal sampling, write $I = \mathbf{1}_{\{K<X<J\}}$ and define the following one-path estimators:

$$Y_V = DF(X), \qquad Y_\Delta = DIX/S,$$
$$Y_\mathcal{V} = DIX(\sqrt{\tau}Z - \sigma\tau), \qquad Y_\mathcal{R} = D\tau\{IX - F(X)\}, \tag{21}$$
$$Y_\Theta = D\left[rF(X) - IX\left(b + \frac{\sigma Z}{2\sqrt{\tau}}\right)\right], \tag{22}$$
$$Y_\Gamma = \frac{DF(X)}{S^2}\left\{\frac{Z^2-1}{v^2} - \frac{Z}{v}\right\}. \tag{23}$$

The first-order Greeks use pathwise differentiation, following the simulation sensitivity framework of Broadie and Glasserman (1996). Gamma uses the second likelihood-ratio weight: differentiating the pathwise indicator as though it were smooth would omit its boundary contribution. I also retain the bounded mixed pathwise/likelihood-ratio rival

$$Y_{\Gamma,\mathrm{mix}} = \frac{DIX}{S^2}\left(\frac{Z}{v} - 1\right). \tag{24}$$

It follows by differentiating delta's fixed-asset density representation, rather than differentiating a random indicator pointwise. Its derivation appears in Appendix C. Boundedness gives a different finite-sample uncertainty guarantee, but does not by itself prove variance dominance over the ordinary likelihood-ratio estimator.

Proposition 4.2 (Unbiased estimators and finite-sample uncertainty)**.** *Each estimator in* (21)–(24) *has expectation equal to its analytic price or Greek and finite second moment. Let*

$$z_- = \frac{\log(K/S) - b\tau}{v}, \quad z_+ = \frac{\log(J/S) - b\tau}{v}, \quad m = \max\{|z_-|, |z_+|\}. \tag{25}$$

Deterministic ranges for price, delta, vega, rho and theta are, respectively,

$$[0, DH], \quad [0, DJ/S], \quad [-B_\mathcal{V}, B_\mathcal{V}], \quad [-D\tau H, D\tau K], \quad [-B_\Theta, B_\Theta],$$
$$B_\mathcal{V} = DJ(\sqrt{\tau}m + \sigma\tau), \qquad B_\Theta = D\left[|r|H + J\left(|b| + \frac{\sigma m}{2\sqrt{\tau}}\right)\right]. \tag{26}$$

For n independent raw paths and a range width R, a distribution-free two-sided confidence radius with failure probability at most α, following Hoeffding (1963), is

$$R\sqrt{\frac{\log(2/\alpha)}{2n}}. \tag{27}$$

For gamma, the bound

$$\mathrm{Var}(Y_\Gamma) \le B_\Gamma^2 := \frac{D^2H^2}{S^4}\left(\frac{2}{v^4} + \frac{1}{v^2}\right) \tag{28}$$

gives the finite-sample Chebyshev radius $B_\Gamma/\sqrt{n\alpha}$. The mixed gamma estimator has range

$$[-B_{\mathrm{mix}}, B_{\mathrm{mix}}], \qquad B_{\mathrm{mix}} = \frac{DJ}{S^2}\left(\frac{m}{v} + 1\right), \tag{29}$$

so it admits the Hoeffding radius with width $2B_{\mathrm{mix}}$. Allocating failure probabilities across outputs and using a union bound gives a simultaneous statement. Sample-variance normal intervals are separate asymptotic diagnostics and do not have these finite-sample coverage guarantees.

Contract	S	H	τ	σ	r	q
Lower kink	100	20	1	0.2	0.03	0.01
Upper kink	120	20	1	0.2	0.03	0.01
Inside spread	110	20	1	0.2	0.03	0.01
Deep cap	160	20	1	0.2	0.03	0.01
Low-volatility short dated	100	20	0.05	0.06	0.03	0.01
Negative rate/dividend	105	20	1	0.25	-0.015	0.045
High volatility	95	30	2	0.65	0.03	0.01

Table 1: Wholly synthetic inputs. Rates, volatility and dividend yields are expressed per year; the cap and spot use common monetary units. The labels describe designed test cases rather than an empirical asset sample.

The numerical study allocates 0.05/49 to each of the six ordinary raw outputs and the additional raw mixed gamma across seven contracts. The corresponding 49-interval family has failure probability at most 0.05 under the ideal independent-normal sampling model. Comparisons between Monte Carlo methods additionally report sample-variance normal intervals as asymptotic diagnostics.

The strongest implemented Monte Carlo comparison combines antithetic normal pairs with a centred discounted-stock control, $DX - Se^{-q\tau}$. I fit each control coefficient on independent pilot pairs and hold it fixed for the main sample. Conditional on the pilot, the controlled mean is unbiased because the control has expectation zero. Each independent pair, rather than its two correlated paths, is the sampling unit. The pilot and both terminal evaluations per pair belong in the computational work. The controlled estimators contain an unbounded lognormal control, so their sample-variance intervals remain asymptotic; the bounded raw-path Hoeffding guarantees are not transferred to them.

5 Synthetic comparison design and numerical evidence

I use seven designed contracts, all with $K = 100$, to exercise both payoff kinks, the interior spread, deep capping, a short low-volatility horizon, negative interest with dividends and a volatile long horizon. Table 1 defines the inputs. No observations are estimated from LSEG or another data provider. The full comparisons and source/code hashes are retained in `results/capped_gbm`.

The coupled meshes use $M = 128, 256, 512, 1024, 2048, 4096, 8192, 16384$ and $N = \lceil M^2/8 \rceil$. Additional experiments refine time at fixed space and space at fixed time. The per-contract heat boxes contain both kinks and use nine diffusion standard deviations of margin, with a minimum margin of 0.3. These finite experiments keep the box fixed; the whole-line convergence theorem additionally requires its tail allowance to vanish. Domain-enlargement checks preserve both h and k, so a boundary change is not confused with changing the spatial resolution.

The spot bump is $\min\{0.1S, S(0.006h^{1/8} + h)\}$. Volatility, rate and maturity bumps are respectively $\min\{0.25\sigma, 0.10\sigma h^{1/4}\}$, $0.01h^{1/4}$ and $\min\{0.25\tau, 0.05\tau h^{1/4}\}$. All actual steps and signed bump biases are retained per comparison. The analytic central difference uses precisely the PDE bump and therefore isolates its bias. The tridiagonal and sine-basis solvers are compared at one, three and 64 time steps, including large values of λ.

Figure 1 exposes non-monotone refinement rather than imposing a fitted rate. Kink/grid alignment and interpolation can affect price error; differencing amplifies these errors in the Greeks. For the short-dated gamma at the finest mesh, the analytic bump bias is approximately -4.28×10^{-4}, while the PDE difference at that bump is approximately 1.45×10^{-5}. The larger gamma discrepancy is therefore mainly a finite-bump effect. Finer price accuracy alone does not remove it. Independent mesh refinements and all five Greek comparisons remain available even

Quantity	Maximum absolute error at $M = 16384$	Units
Price	9.950×10^{-7}	Monetary units
Delta	1.111×10^{-4}	Per unit spot
Gamma	4.137×10^{-4}	Per unit spot squared
Vega	1.829×10^{-3}	Per unit volatility
Rho	3.171×10^{-4}	Per unit rate
Theta	1.026×10^{-3}	Per year of calendar time

Table 2: Worst absolute discrepancy from the analytic price/Greek across the seven synthetic contracts. Greek errors include the declared bump bias. Different derivative units are not combined into a single accuracy score.

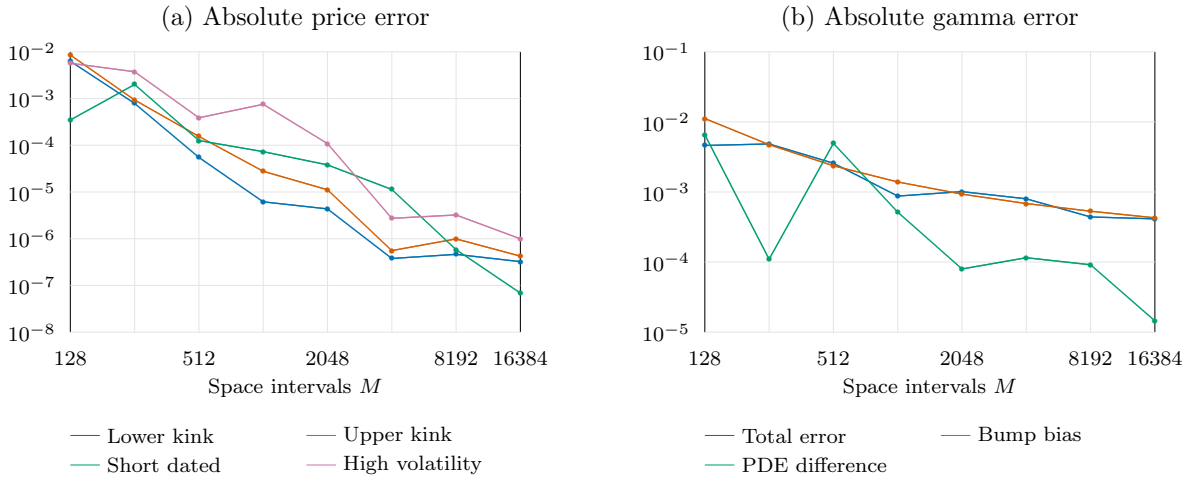


Figure 1: Synthetic convergence evidence. Panel (a) contrasts the lower and upper payoff kinks, a short-dated low-volatility claim and a high-volatility claim. Panel (b) separates the short-dated gamma total error into analytic bump bias and the PDE difference at the same bump. Absolute component errors need not add because their signed contributions can cancel. Both axes are logarithmic. All seven contracts and all five Greeks are retained in the comparison files.

where a selected plot would conceal adverse behaviour.

Every Monte Carlo method receives 262,144 terminal-payoff evaluations per contract. Raw sampling uses that many independent normals. Antithetic sampling uses 131,072 independent pairs. Controlled sampling uses 16,384 independent pilot pairs and 114,688 production pairs, charging both terminal evaluations of each pair. A control coefficient is fitted separately for each output and clipped to $[-10^6, 10^6]$; the pilot is independent of production. Shared production normals between methods are declared, and no confidence claim is made for their paired differences. The seed is 732451, with deterministic case offsets. Terminal-evaluation budgets measure sampling work; arithmetic costs and wall-clock time are not asserted to be identical between estimators.

The 49 raw outputs comprise the price, five Greeks and additional mixed gamma in each of seven contracts. Each receives failure allowance 0.05/49. Their simultaneous finite-sample bounds apply to ideal independent exact normal draws and exact arithmetic. Their floating-point constants and pseudo-random normal generation have not been formally certified. Normal sample-variance intervals are reported separately as pointwise asymptotic diagnostics. Observed inclusion in the conservative bounds is a sanity check, not a proof of coverage.

The bounded mixed estimator reduces measured raw gamma variance in every tested contract. This is an observed comparison, not uniform dominance over model parameters. Con-

Contract	Analytic Γ	Raw LR SE	Raw mixed SE	Variance ratio
Lower kink	4.026×10^{-3}	7.689×10^{-5}	1.966×10^{-5}	15.293
Upper kink	-7.259×10^{-3}	6.210×10^{-5}	2.269×10^{-5}	7.493
Inside spread	-3.183×10^{-3}	7.019×10^{-5}	1.523×10^{-5}	21.240
Deep cap	-2.747×10^{-3}	4.014×10^{-5}	2.021×10^{-5}	3.945
Low-volatility short dated	0.296226	3.320×10^{-3}	8.632×10^{-4}	14.792
Negative rate/dividend	2.713×10^{-3}	4.478×10^{-5}	1.272×10^{-5}	12.400
High volatility	-3.680×10^{-4}	4.251×10^{-6}	2.287×10^{-6}	3.455

Table 3: Raw gamma estimators at equal terminal-evaluation budgets. SE is the measured standard error, not a finite-sample confidence radius. The variance ratio is ordinary payoff-LR estimator variance divided by mixed-estimator variance. All antithetic and independently fitted control variants remain in the full comparison.

trolled ordinary likelihood-ratio gamma remains more efficient in several cases. Stock controls also worsen measured delta and rho efficiency for the interior-spread contract at the matched total budget. The analytic call spread has neither sampling error nor a finite-domain or mesh error and remains the strongest baseline here.

6 Interpretation and reproducibility

This companion closes a specific preparation gap by connecting an admissible financial strategy, a PDE, a nonsmooth numerical convergence proof and independently checked sensitivities. The bounded mixed-gamma derivation gives a useful estimator and a finite-sample uncertainty statement for this claim. It is an explicit specialisation of sensitivity principles, not a claimed new quantum theory or a completed priority claim. The comparison retains negative gamma/vega, non-monotone errors and adverse control rankings where they occur.

The study checks a stipulated one-asset model with known parameters. It does not calibrate a financial model, validate market prices, solve a general portfolio/control problem or establish a quantum advantage. It also does not extend a one-asset analytic benchmark into a closed-form solution for arbitrary baskets. Those boundaries are essential when using the companion as evidence of financial mathematics and computation.

The engine is `capped_gbm.py`; `run_capped_gbm_study.py` reproduces the comparisons. The tests include independent active-region Gaussian quadrature of the mixed gamma, analytic sensitivity checks and the direct/sine-basis backward-Euler comparison. The source ledger retains the final published versions and precise sections consulted. Detailed arguments follow in the appendices. I retain this document as a completed companion study; the main quantum basket paper remains a working paper.

A Financial and analytic derivations

A.1 Change of measure and the self-financing equation

For a concrete construction, suppose under a physical measure P that $dS_t = (\mu - q)S_t\,dt + \sigma S_t\,dW_t^P$ with constant total-return drift μ. Set $\vartheta = (\mu - r)/\sigma$. Novikov's condition follows from $\exp(\vartheta^2 T/2) < \infty$ on every finite horizon. Thus

$$\left.\frac{dQ}{dP}\right|_{\mathcal{F}_T} = \exp\{-\vartheta W_T^P - \vartheta^2 T/2\}, \qquad W_t^Q = W_t^P + \vartheta t \tag{30}$$

defines an equivalent probability measure and yields (2). The stochastic exponential is strictly positive and has expectation one by the Gaussian moment-generating function. For this constant

coefficient, the Brownian conclusion can also be checked directly. On any deterministic partition, the density is the product of factors $\exp\{-\vartheta\Delta W^P - \vartheta^2\Delta t/2\}$. Multiplying the independent $N(0,\Delta t)$ increment densities by these factors and completing the square changes each mean to $-\vartheta\Delta t$, leaves its variance Δt and preserves their factorisation. Thus the increments of $W^Q = W^P + \vartheta t$ are independent centred normals with the Brownian variances. The same calculation conditional on the preceding Brownian filtration proves independence from its past. Continuity is inherited from W^P. Hence W^Q is a Brownian motion under Q, an elementary verification of this established constant-coefficient measure change. Consequently this construction does not depend on an unverified calibration of μ.

Let $B_t = e^{rt}$, let δ_t be the share holding and let γ_t be the bank-account holding. A self-financing wealth $P_t = \delta_t S_t + \gamma_t B_t$ receives asset dividends, so

$$dP_t = \delta_t\, dS_t + q\delta_t S_t\, dt + r(P_t - \delta_t S_t)\, dt = rP_t\, dt + \delta_t\sigma S_t\, dW_t^Q. \tag{31}$$

For an admissible strategy, the share holding is predictable and

$$\int_0^T e^{-2rt}\delta_t^2\sigma^2 S_t^2\, dt < \infty \quad Q\text{-almost surely}, \tag{32}$$

while discounted wealth is bounded below by a deterministic constant. For such a strategy, discounted wealth is a local martingale bounded below and hence a supermartingale. One may see the latter implication by localisation and conditional Fatou's lemma after adding the fixed lower bound. An arbitrage with zero initial wealth and non-negative, sometimes strictly positive terminal wealth would contradict the supermartingale expectation inequality. This proves no arbitrage within this explicit admissible strategy class; it does not assert a general fundamental theorem for arbitrary markets.

A.2 The analytic call and replication

The event $X > A$ is $Z > -d_{2,A}$. Completing the square gives

$$\mathbb{E}[e^{vZ}\mathbf{1}_{\{Z>z\}}] = e^{v^2/2}\Phi(v - z). \tag{33}$$

Substitution in $D\mathbb{E}[(X - A)^+]$ yields (5); payoff linearity yields the call spread. The normal identity

$$Se^{-q\tau}\varphi(d_{1,A}) = AD\varphi(d_{2,A}) \tag{34}$$

follows directly by subtracting the two squared normal arguments. It cancels the derivative-of-Φ terms when differentiating the call formula. The normal arguments have derivatives

$$\begin{aligned}
&\partial_S d_{1,A} = \partial_S d_{2,A} = \frac{1}{Sv}, && \\
&\partial_\sigma d_{1,A} = \sqrt{\tau} - \frac{d_{1,A}}{\sigma}, && \partial_\sigma d_{2,A} = -\frac{d_{1,A}}{\sigma}, \\
&\partial_r d_{1,A} = \partial_r d_{2,A} = \frac{\sqrt{\tau}}{\sigma}, && \\
&\partial_\tau d_{1,A} = \frac{r - q + \sigma^2/2}{\sigma\sqrt{\tau}} - \frac{d_{1,A}}{2\tau}, && \partial_\tau d_{2,A} = \partial_\tau d_{1,A} - \frac{\sigma}{2\sqrt{\tau}}.
\end{aligned} \tag{35}$$

In the spot derivative of C_A, the two density terms cancel exactly, leaving $e^{-q\tau}\Phi(d_{1,A})$; another spot derivative gives $e^{-q\tau}\varphi(d_{1,A})/(Sv)$. In the volatility derivative, their combined contribution is $Se^{-q\tau}\varphi(d_{1,A})(\partial_\sigma d_{1,A} - \partial_\sigma d_{2,A}) = Se^{-q\tau}\varphi(d_{1,A})\sqrt{\tau}$. The rate-argument derivatives cancel, leaving $\tau AD\Phi(d_{2,A})$ from differentiating the discount factor. Finally,

$$\partial_\tau C_A = -qSe^{-q\tau}\Phi(d_{1,A}) + rAD\Phi(d_{2,A}) + \frac{Se^{-q\tau}\sigma\varphi(d_{1,A})}{2\sqrt{\tau}}. \tag{36}$$

Subtracting the corresponding strike-J expressions from those at strike K proves every formula in (6)–(10). The lognormal convolution is smooth at positive maturity, even though F has two kinks. Dominated convergence proves continuity at expiry: $X \to S$ almost surely, F is continuous and $0 \le F \le H$.

To verify the heat transformation explicitly, first write $x = \log(S/K)$ and $V = e^{-r\tau}u(\tau, x)$. The chain rule gives $SV_S = e^{-r\tau}u_x$ and $S^2V_{SS} = e^{-r\tau}(u_{xx} - u_x)$. Equation (4) therefore reduces to $u_\tau = au_{xx} + bu_x$. Setting $u(\tau, x) = w(\tau, x + b\tau)$ gives $w_\tau = aw_{yy}$. Conversely, Gaussian convolution proves (11); reversing the calculation proves (4) without assuming it in the valuation argument.

With calendar time t and expiry T, apply Itô's formula to $V(S_t, T - t)$ on $[0, T - \varepsilon]$. The pricing equation gives

$$dV(S_t, T - t) = rV(S_t, T - t)\,dt + \sigma S_t V_S(S_t, T - t)\,dW_t^Q. \tag{37}$$

This agrees with (31) for $\delta = V_S$ and $\gamma = (V - SV_S)/B$. Furthermore $0 \le V_S \le e^{-q(T-t)}$ by (6), and the Gaussian moment-generating function gives the explicit second moment

$$\mathbb{E}_Q S_t^2 = S_0^2 \exp\{[2(r - q) + \sigma^2]t\}. \tag{38}$$

It is finite and bounded on every finite horizon. Hence $\mathbb{E}_Q \int_0^T e^{-2rt}\sigma^2 S_t^2 V_S^2\,dt < \infty$. The discounted stochastic gain is therefore a true square-integrable martingale. Continuity at expiry gives terminal wealth $F(S_T)$ as $\varepsilon \downarrow 0$. Its discounted wealth is non-negative, so the strategy is admissible. Any bounded quoted claim valuation different from the replication cost permits buying the cheaper position, selling the dearer one and investing the initial difference; their terminal claim cash flows cancel. This gives a strictly positive terminal gain with zero initial wealth and a bounded-below discounted wealth process, contradicting no arbitrage. Any other bounded classical solution with this terminal payoff has a discounted martingale representation with the same conditional terminal expectation; equivalently, apply the stopped Itô argument and bounded convergence. Its price equals (3). This proves the stated uniqueness.

B Numerical stability and convergence proofs

B.1 The matrix maximum principle

Let A_h be the $(M - 1) \times (M - 1)$ matrix with diagonal $1 + 2\lambda$ and off-diagonals $-\lambda$. If $A_h z \ge 0$ componentwise and z had a negative minimum at an interior row, that row would satisfy $(A_h z)_j \le z_j < 0$. At an edge row, the missing neighbour is interpreted as zero and gives an even stronger contradiction. Thus $z \ge 0$. The same argument applied to a null vector and its negative proves that A_h is nonsingular; consequently $A_h^{-1} \ge 0$. The row sums of A_h are one in interior rows and $1 + \lambda$ at the two edge rows. Therefore $A_h\mathbf{1} \ge \mathbf{1}$, whence

$$0 \le A_h^{-1}\mathbf{1} \le \mathbf{1}, \qquad \|A_h^{-1}\|_\infty \le 1. \tag{39}$$

This proves contraction for two solutions with equal boundary values. For positivity and the upper bound, the actual right-hand side is the previous interior vector plus λH at the last row. If the previous vector lies between 0 and $H\mathbf{1}$, this right-hand side lies between 0 and $A_h(H\mathbf{1})$. Applying the non-negative inverse proves the invariant interval by induction. No condition relating k and h is used.

The discrete eigenvectors are $\sin(j\ell\pi/M)$ and substitution in the centred second difference gives $-\nu_\ell$. The affine lift $\ell(y) = H(y - L)/(U - L)$ has zero discrete second difference. Each lifted mode is consequently multiplied by $(1 + ak\nu_\ell)^{-1}$ per step, which proves (14) after N steps. This identity is in exact arithmetic; the direct-step comparison checks implementation error.

B.2 Mollification without a false smooth-payoff assumption

Let $\rho_\eta(z) = \eta^{-1}\rho(z/\eta)$ and $f_\eta = f * \rho_\eta$. Global Lipschitz continuity gives

$$\|f_\eta - f\|_\infty \le L_f m_1 \eta, \qquad \|f_\eta^{(4)}\|_\infty \le L_f \|\rho_\eta'''\|_1 = L_f c_3 \eta^{-3}. \tag{40}$$

The second estimate uses the weak first derivative f', which exists almost everywhere and is bounded by L_f. Moving the three remaining derivatives to the smooth kernel proves the formula and its bound. Since the radius is smaller than the distance of either kink to the boundary, f_η equals 0 near L and H near U.

Subtract the affine lift from f_η and shift the domain to $[0, U-L]$. The lifted function is zero at both endpoints and affine in their neighbourhoods. Its even derivatives, of order at least two, vanish there. Odd reflection at both endpoints therefore produces a smooth function of period $2(U-L)$: the value and every derivative agree at each reflection. In particular, the fourth derivative of the extension has maximum norm at most $L_f c_3 \eta^{-3}$. Convolve this extension with the whole-line Gaussian heat kernel and add the affine lift back. Oddness keeps the lifted solution zero at both boundaries, so this is precisely the finite-domain smooth solution u_η. Gaussian convolution contracts the maximum norm and commutes with derivatives; hence

$$\sup_{0\le s\le\tau} \|\partial_y^4 u_\eta(s,\cdot)\|_\infty \le L_f c_3 \eta^{-3}, \qquad \partial_s^2 u_\eta = a^2 \partial_y^4 u_\eta. \tag{41}$$

At time $s_n = nk$, insert $u_\eta(s_n, y_j)$ into the numerical equation and divide its residual by k. The integral remainder for backward time differencing is at most $k \sup|\partial_s^2 u_\eta|/2$. The integral or Taylor remainder for the centred second difference is at most $h^2 \sup|\partial_y^4 u_\eta|/12$. Thus the residual R_j^n obeys

$$|R_j^n| \le \left(\frac{a^2 k}{2} + \frac{a h^2}{12}\right) L_f c_3 \eta^{-3}. \tag{42}$$

If the smooth grid starts from f_η, its error satisfies $A_h e^n = e^{n-1} - kR^n$ with $e^0 = 0$. Equation (39) and induction give $\|e^N\|_\infty \le Nk \sup_n \|R^n\|_\infty$. Both the continuous maximum principle and the discrete contraction bound the change from f_η back to f by $L_f m_1 \eta$. Adding these two changes proves (16). Taking η proportional to $(h^2 + k)^{1/4}$, eventually below the fixed boundary margin, proves convergence and the stated upper rate. This is a proof for the actual kinked initial condition, not for a replacement smooth contract.

B.3 Domain truncation and interpolation

Let $Y_s = x + \sigma B_s$, and let ζ be its first exit from $[L, U]$. The stopped Brownian representation of the finite-domain solution is

$$w^{\text{box}}(\tau, x) = \mathbb{E}_x[f(Y_\tau)\mathbf{1}_{\{\zeta>\tau\}}] + \mathbb{E}_x[g(Y_\zeta)\mathbf{1}_{\{\zeta\le\tau\}}], \qquad g(L) = 0, \quad g(U) = H. \tag{43}$$

It follows either from the stopped Itô formula for positive-time smooth solutions and bounded convergence, or from the killed heat semigroup. The full solution is $\mathbb{E}_x f(Y_\tau)$. On exit, the terminal payoff and boundary value are both in $[0, H]$, so their difference is bounded in absolute value by H. The reflection principle bounds each one-sided exit event by twice its corresponding normal tail; a union bound proves (17). The upper and lower exit contributions can have opposite signs, so this is an absolute error bound rather than an ordering of the two prices.

At positive maturity, differentiate the whole-line convolution once using the bounded weak derivative f', then once using the Gaussian density. For a Gaussian with standard deviation v, the L^1 norm of its density derivative is $\sqrt{2/\pi}/v$. Therefore

$$\|w_{yy}(\tau,\cdot)\|_\infty \le \frac{L_f}{\sigma\sqrt{\tau}}\sqrt{\frac{2}{\pi}}. \tag{44}$$

Linear interpolation on an interval of width h has error at most $h^2 \sup |w_{yy}|/8$. Interpolation weights are non-negative and sum to one. Apply the domain bound at the two bracketing nodes and the nodal discretisation bound to these same weights; add the whole-line interpolation error and discount. This proves (18), without requiring a new boundary regularity bound for w^{box}.

B.4 Uniform consistency for Greeks

For a central first difference, price error contributes at most $(E + E)/(2d) = E/d$. Taylor's theorem with an integral remainder gives bias at most $M_3 d^2/6$. For the central second difference, price error contributes at most $(E + 2E + E)/d^2 = 4E/d^2$, and the fourth-order remainder is at most $M_4 d^2/12$. These prove (19) and (20).

On a compact parameter set with positive lower bounds for S, σ and τ, the analytic formula is smooth and its third and fourth parameter derivatives are bounded. Its transformed queries also lie in a compact interval. Let $L \to -\infty$ and $U \to \infty$, keep $h \to 0$ and $k \to 0$, and use a radius proportional to $(h^2 + k)^{1/4}$. The Gaussian exit tails vanish uniformly because $\sigma\sqrt{\tau}$ has a finite upper bound, and the other terms in (18) vanish uniformly. The discount factors are bounded even when r is negative. Thus $E_n \to 0$ uniformly. Choosing the bump sequence as stated in Proposition 4.1 proves Greek convergence. A fixed domain or a fixed price error does not meet this argument merely because its bumps are made smaller.

C Monte Carlo derivations and uncertainty proofs

C.1 Pathwise first derivatives

The capped payoff is Lipschitz in its asset argument, with derivative $\mathbf{1}_{\{K<x<J\}}$ except at two points. The continuously distributed X hits those points with probability zero. Its parameter derivatives are

$$X_S = X/S, \qquad X_\sigma = X(\sqrt{\tau}Z - \sigma\tau), \qquad X_r = \tau X, \qquad X_\tau = X\left(b + \frac{\sigma Z}{2\sqrt{\tau}}\right). \tag{45}$$

On any compact parameter neighbourhood with $S, \sigma, \tau > 0$, difference quotients of the Lipschitz payoff are dominated by a constant times $(1 + |Z|)\exp(C|Z|)$, which is integrable against a Gaussian. This follows from the mean-value theorem for the smooth terminal map X and the Lipschitz bound for F. Dominated convergence therefore justifies interchanging differentiation and expectation. Differentiating the discount factor as well gives (21) and (22). Neither payoff kink contributes a first-order term because the payoff itself is continuous there.

On the active region, $X \le J$ and $Z \in (z_-, z_+)$, giving the delta, vega and theta ranges in (26). For rho the expression $IX - F(X)$ is exactly 0, K or $-H$ in the below-strike, active and capped regions, respectively. This proves its stated tighter range.

C.2 The likelihood-ratio gamma

For $x > 0$, the lognormal density is

$$p_S(x) = \frac{1}{xv\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{\log(x/S) - b\tau}{v}\right)^2\right]. \tag{46}$$

Writing $z = (\log(x/S) - b\tau)/v$, direct differentiation gives

$$\frac{\partial_S p_S(x)}{p_S(x)} = \frac{z}{Sv}, \qquad \frac{\partial_S^2 p_S(x)}{p_S(x)} = \frac{1}{S^2}\left(\frac{z^2 - 1}{v^2} - \frac{z}{v}\right). \tag{47}$$

On a compact spot neighbourhood away from zero, the density and its first two spot derivatives admit a common envelope

$$C(1+|\log x|^2)x^{-1}\exp\{-|\log x|^2/(4v^2)\}, \qquad x>0, \tag{48}$$

for a finite constant C. Indeed $\log S + b\tau$ is bounded on the neighbourhood and $(u-m)^2 \geq u^2/2 - m^2$; the derivative weights are polynomials of degree at most two in $u = \log x$, with bounded inverse spot factors. The envelope is integrable by the substitution $u = \log x$, and remains integrable after multiplication by $F \leq H$. Two applications of dominated convergence to $D\int F(x)p_S(x)\,dx$ prove unbiasedness of (23). Using $F \leq H$ gives

$$\begin{aligned}\mathbb{E}[Y_\Gamma^2] &\leq \frac{D^2H^2}{S^4}\mathbb{E}\left[\left(\frac{Z^2-1}{v^2}-\frac{Z}{v}\right)^2\right] \\ &= \frac{D^2H^2}{S^4}\left(\frac{2}{v^4}+\frac{1}{v^2}\right),\end{aligned} \tag{49}$$

because $\mathbb{E}(Z^2-1)^2 = 2$, $\mathbb{E}Z^2 = 1$ and the odd mixed moment is zero. Variance is no larger than the second moment, proving (28). Gamma's likelihood-ratio estimator is unbounded; a bounded-payoff concentration inequality cannot be applied to it without this separate argument.

C.3 A bounded mixed gamma rival

The pathwise delta has the fixed-asset representation

$$\Delta = \frac{D}{S}\int_K^J xp_S(x)\,dx. \tag{50}$$

The integration limits are fixed when spot varies. On this bounded interval away from zero, $x\partial_S p_S(x)$ has a common integrable envelope on every compact spot neighbourhood. Differentiation is therefore legitimate even though the payoff derivative is an indicator. Using (47),

$$\begin{aligned}\partial_S\Delta &= -\frac{D}{S^2}\int_K^J xp_S(x)\,dx + \frac{D}{S}\int_K^J x\partial_S p_S(x)\,dx \\ &= \frac{D}{S^2}\int_K^J xp_S(x)\left(\frac{z}{v}-1\right)\,dx.\end{aligned} \tag{51}$$

This proves unbiasedness of (24). It includes the boundary effect through the density derivative. On the active region, $X < J$ and $|Z| \leq m$, proving (29) and its finite second moment. As an independent analytic check, Gaussian tilting reduces its expectation to $e^{-q\tau}\{\varphi(d_{1,K})-\varphi(d_{1,J})\}/(Sv)$, in agreement with (7). I combine the established pathwise and likelihood-ratio constructions of Broadie and Glasserman (1996), deriving the capped-payoff expression explicitly here; I make no priority claim. The ordinary and mixed gamma estimators are compared using identical financial parameters and matched terminal evaluation budgets; neither estimator is discarded after seeing its variance.

C.4 Confidence guarantees and stronger classical sampling

For completeness, the bounded-variable concentration argument of Hoeffding (1963) can be checked directly. If Y lies in an interval of width $R > 0$, define $A(s) = \log\mathbb{E}\exp\{s(Y-\mathbb{E}Y)\}$ for real s. Boundedness justifies differentiating twice. The second derivative is the variance of Y under the exponentially tilted probability law. That law has the same interval support; centring at its midpoint gives $A''(s) \leq R^2/4$. Since $A(0) = A'(0) = 0$, integration twice yields

$$A(s) \leq R^2s^2/8, \qquad s\in\mathbb{R}. \tag{52}$$

For independent observations and $s > 0$, exponential Markov's inequality and the product moment-generating function therefore give

$$\mathbb{P}(\overline{Y} - \mathbb{E}Y \geq t) \leq \exp\{n(R^2 s^2/8 - st)\}. \tag{53}$$

The minimiser is $s = 4t/R^2$, giving the upper-tail bound $\exp(-2nt^2/R^2)$. Apply the same argument to $-Y$ and sum the two tail bounds. Solving $2\exp(-2nt^2/R^2) = \alpha$ proves (27). If $R = 0$, the observation is constant and its confidence radius is zero. Independence gives $\operatorname{Var}(\overline{Y}_\Gamma) \leq B_\Gamma^2/n$; Chebyshev's inequality gives the stated gamma radius. A union bound requires no independence between different outputs calculated on the same paths. Their allocated failure probabilities must sum to the intended familywise level.

For the stock control $C = DX - Se^{-q\tau}$, the Gaussian exponential moment gives $\mathbb{E}C = 0$ and a finite second moment. For any estimator Y, a deterministic coefficient β yields

$$\operatorname{Var}(Y - \beta C) = \operatorname{Var}(Y) - 2\beta \operatorname{Cov}(Y, C) + \beta^2 \operatorname{Var}(C). \tag{54}$$

When $\operatorname{Var}(C) > 0$, the population minimiser is $\beta_* = \operatorname{Cov}(Y, C)/\operatorname{Var}(C)$ and the minimum variance is $\operatorname{Var}(Y) - \operatorname{Cov}(Y, C)^2/\operatorname{Var}(C)$. An estimated pilot coefficient need not attain that minimum. Independence of pilot and main sample preserves conditional unbiasedness; a zero pilot denominator is handled by selecting zero control coefficient. Antithetic pairs replace $Y(Z)$ and $C(Z)$ with their averages at Z and $-Z$ and preserve their expectations, but variance reduction for every Greek is not guaranteed merely by symmetry. The comparison therefore reports measured variance and complete sampling work. For fixed finite pilot coefficients, the main-pair estimators have finite variance, so their studentised normal intervals have asymptotic validity under non-zero variance. Finite-sample exact coverage is not claimed.

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